SEARCHING AND SORTING

Insertion Sort - Part 1

```
#include <stdio.h>
void print(int ar_size, int* ar) {
    int i;
    for(i=0; i<ar_size; i++) {
        printf("%d ", ar[i]);
    }
    printf("\n");
}

#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <assert.h>

/* Head ends here */
void insertionSort(int ar_size, int *  ar) {
    int j = ar_size-1;
    int v = ar[j];
    while(v < ar[j-1]) {
        ar[j] = ar[j-1];
        j--;
        print(ar_size, ar);
    }
    ar[j] = v;
    print(ar_size, ar);
```

```
}

/* Tail starts here */
int main() {

   int _ar_size;
scanf("%d", &_ar_size);
int _ar[_ar_size], _ar_i;
for(_ar_i = 0; _ar_i < _ar_size; _ar_i++) {
   scanf("%d", &_ar[_ar_i]);
}

insertionSort(_ar_size, _ar);

   return 0;
}
```

Insertion Sort - Part 2

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
#include <assert.h>
/* Head ends here */
void insertionSort(int ar_size, int *  ar) {
    for (int i = 1; i < ar_size; ++i) {
        int j = i - 1;
        int p = ar[i];
        while (j >= 0 && p < ar[j]) {
            ar[j+1] = ar[j];
            j--;
        }
        ar[j+1] = p;
        printf("%d", ar[0]);
        for (int k = 1; k < ar_size; ++k) {
            printf(" %d", ar[k]);
        }
        printf("\n");
    }
}
/* Tail starts here */
int main() {
   int _ar_size;
scanf("%d", &_ar_size);
int _ar[_ar_size], _ar_i;
for(_ar_i = 0; _ar_i < _ar_size; _ar_i++) {
```

```
    scanf("%d", &_ar[_ar_i]);
}

insertionSort(_ar_size, _ar);

    return 0;
}
```

Correctness and the Loop Invariant

```
#include <stdio.h>
#include <string.h>
```

```
#include <math.h>
#include <stdlib.h>
#include <assert.h>
/* Head ends here */
#include <stddef.h>
void insertionSort(int ar_size, int *  ar) {
    int i,j;
    int value;
    for(i=1;i<ar_size;i++)
    {
        value=ar[i];
        j=i-1;
        while(j>=0 && value<ar[j])
        {
            ar[j+1]=ar[j];
            j=j-1;
        }
        ar[j+1]=value;
    }
    for(j=0;j<ar_size;j++)
        {
            printf("%d",ar[j]);
            printf(" ");
        }
}
/* Tail starts here */
int main(void) {
   int _ar_size;
```

```
scanf("%d", &_ar_size);
int _ar[_ar_size], _ar_i;
for(_ar_i = 0; _ar_i < _ar_size; _ar_i++) {
   scanf("%d", &_ar[_ar_i]);
}

insertionSort(_ar_size, _ar);

   return 0;
}
```

Running Time of Algorithms

```
#include <stdio.h>
#include <string.h>
#include <math.h>
```

```
#include <stdlib.h>
#include <assert.h>

/* Head ends here */
void insertionSort(int ar_size, int *  ar,int *shifts) {
int temp=ar[ar_size-1],i;
    for(i=ar_size-2;i>=0;i--)
    {
        if(ar[i]>temp){
            ar[i+1]=ar[i];
            *shifts=*shifts+1;
            }
        else
            break;

    }
    ar[i+1]=temp;



}

/* Tail starts here */
int main() {

   int _ar_size,i,j,shifts=0;
scanf("%d", &_ar_size);
int _ar[_ar_size], _ar_i;
```

```
for(_ar_i = 0; _ar_i < _ar_size; _ar_i++) {
   scanf("%d", &_ar[_ar_i]);
}
for(i=2;i<=_ar_size;i++)
{
    insertionSort(i, _ar,&shifts);
}
    printf("%d",shifts);
   return 0;
}
```

Counting Sort 1

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>

int main() {

    int n,i;
    int b[100],a;
    scanf("%d",&n);

    for(i=0;i<100;i++)
    {
        b[i]=0;
    }
```

```
    for(i=0;i<n;i++)
    {
        scanf("%d",&a);
         b[a]++;
    }
    for(i=0;i<100;i++)
    {
        printf("%d ", b[i]);
    }

    return 0;
}
```

RECURSION AND BIT MANIPULATION

Crossword Puzzle

The Power Sum

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>

int the_power_sum(int n, int m,int p){
    int tmp = pow(m,p);
    if(tmp == n) return 1;
    if(tmp > n) return 0;
 return the_power_sum(n,m+1,p) + the_power_sum(n-tmp, m+1,p);
}
int main() {
    int n,p;
    scanf("%d%d",&n,&p);
    printf("%d", the_power_sum(n,1,p));

    return 0;
}
```

Counter Game

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
int isPow2(long unsigned  int);
unsigned long int largePow(long unsigned int);
```

```
int main() {

    int t,i,win;
   long unsigned int n;
    scanf("%d",&t);
   for(i=0;i<t;++i)
       {
       win=0;
       scanf("%lu",&n);
       if(n==1)
           printf("Richard\n");
       else
           {
           while(n!=1)
               {
               if(isPow2(n))
                   n>>=1;
               else
                   n-=largePow(n);
               ++win;
           }
       }
       if(win%2==0)
           printf("Richard\n");
       else
           printf("Louise\n");
   }
   return 0;
```

```
}
int isPow2(long unsigned int n)
    {
    return !(n&(n-1));
}
long unsigned int largePow(long unsigned int n)
    {
    long unsigned int m;
    while(n)
        {
        m=n;
        n=n&(n-1);
    }
    return m;
}
```

GREEDY AND DYNAMIC PROGRAMMING

The Coin Change Problem

Sherlock and Cost

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
```

```
#include <stdbool.h>

int main() {
    int T,N,B,L,R,ML,MR,X,Y,P,Q;
    scanf("%d",&T);
    for(int i = 0; i < T; i++) {
        scanf("%d",&N);
        for(int j = 0; j < N; j++) {
            scanf("%d",&B);
            if(j) {
                X = L - 1 + ML;
                Y = R - 1 + MR;
                P = abs(L - B) + ML;
                Q = abs(R - B) + MR;
                ML = (X > Y ? X : Y);
                MR = (P > Q ? P : Q);
            } else {
                ML = MR = 0;
            }
            L = 1;
            R = B;
        }
        printf("%d\n", (ML > MR ? ML : MR));
    }
    return 0;
}
```

Marc's Cakewalk

```
#include <math.h>

#include <stdio.h>

#include <string.h>

#include <stdlib.h>

#include <assert.h>

#include <limits.h>

#include <stdbool.h>

void swap(int *a,int *b)
```

```
{
    int temp;
    temp = *a;
    *a = *b;
    *b=temp;
}
int partition(int *x,int start,int end)
{
    int pivot,pindex,i;
    pivot = x[end];
    pindex = start;
    for(i=start;i<end;i++)
    {
        if(x[i]>=pivot)
         {
           swap(&x[i],&x[pindex]);
           pindex = pindex + 1;
         }
    }
    swap(&x[pindex],&x[end]);
    return pindex;
}
void quicksort(int *x,int start,int end)
{
    if(start<end)
   {
     int i = partition(x,start,end);
      quicksort(x,start,i-1);
```

```
      quicksort(x,i+1,end);
    }
}
int main(){
    int n, calories_i, *calories;
    int i;
    int sum = 0;
    scanf("%d",&n);
    calories = malloc(sizeof(int) * n);
    for(calories_i = 0; calories_i < n; calories_i++)
    {
       scanf("%d",&calories[calories_i]);
    }
    quicksort(calories,0,n-1);

    for(i=0;i<n;i++)
    {
        sum += calories[i]*((int)pow(2,i));
    }
    printf("%d",sum);
    // your code goes here
    return 0;
}
```

STRINGS

Strong Password

```
#include <math.h>
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <assert.h>
#include <limits.h>
#include <stdbool.h>
```

```
int minimumNumber(int n, char* password) {

    int L_c = 0;
    int U_c = 0;
    int no = 0;
    int s_c = 0;

    int len = strlen(password);
    int i;
    for(i = 0; i<n; i++)
    {
        if(password[i] >= 'a' && password[i] <= 'z')
            L_c++;
        else if(password[i] >= 'A' && password[i] <= 'Z')
            U_c++;
        else if(password[i] >= '0' && password[i] <= '9')
            no++;
        else
            s_c++;
    }

    int add = 0;
    if(L_c == 0)
        add++;
    if(U_c == 0)
        add++;
    if(no == 0)
        add++;
```

```
    if(s_c == 0)
        add++;

    len = len + add;

    if(len < 6)
        add = add + 6 - len;

    return add;
}

int main() {
    int n;
    scanf("%i", &n);
    char* password = (char *)malloc(512000 * sizeof(char));
    scanf("%s", password);
    int answer = minimumNumber(n, password);
    printf("%d\n", answer);
    return 0;
}
```

Caesar Cipher

```
#include <stdio.h>
#include <string.h>
#include <math.h>
#include <stdlib.h>
```

```
int main() {

    int n,i,j,k;
    char ar[101];
    unsigned char x;
    scanf("%d",&n);
    scanf("%s",ar);
    scanf("%d",&k);
    for(i=0;i<n;i++)
    {
        x=ar[i];
        if(x>=97 && x<=122)
        {
            x=x+(k%26);
            if(x>122)
            {
                x=96+(x-122);
            }
            ar[i]=x;
        }
        else if(x>=65 && x<=90)
        {
            x=x+(k%26);
            if(x>90)
            {
                x=64+(x-90);
            }
```

```
            ar[i]=x;
        }
    }
    printf("%s",ar);
    return 0;
}
```

Pangrams

```
#include <stdio.h>
char s[10000];
int main()
    {
    gets(s);
    int f[300]={0},ans=0,i;
    int l=strlen(s);
    for(i=0;i<l;i++)
        {
        if(s[i]==' ')
            continue;
        if(s[i]>='a')
```

```
            s[i]=s[i]-('a'-'A');
        if(!(f[s[i]]++))
            ans++;

    }
     if(ans!=26)
         printf("not ");
     printf("pangram\n");
    return 0;
}
```

RANGE QUERIES

Prefix Sum Array

Populate a Prefix-Sum Array from an array of integer elements.

Also create a function to calculate the sum of elements between a given range using the Prefix-Sum Array.

```
#include<stdio.h>
void display(int arr[], int n)
{
        int i;
        for (i=0;i<n;i++)
        {
                printf("\t %d ",arr[i]);
        }
```

```
}

create_prefix_sum_array(int arr[], int n)
{
	int i;
	for (i = 1; i<n; i++)
	{
		arr[i] = arr[i] + arr[i-1];
	}
}

void sum(int arr[], int n, int a, int b){
	int result;
	if (a == 0)
		result = arr[b];
	else
		result = arr[b] - arr[a-1];
	printf("The sum is %d", result);
}

int main(){
	int n, i, arr[10], a, b;
	printf("enter the number of elements\n");
	scanf("%d",&n);
	printf("enter the array elements\n");
	for (i=0;i<n;i++)
	{
		scanf("%d",&arr[i]);
```

```
    }

    printf("\n Original array");
    display(arr, n);

    create_prefix_sum_array(arr, n);

    printf("\n Prefix sum array");
    display(arr,n);

    printf("\n enter the range to find the sum");
    scanf("%d%d",&a,&b);
    sum(arr, n, a, b);
    return 0;
}
```

Fenwick Tree Construction

Populate a Fenwick Tree from an array of integer elements.

Given the value of a node in a Fenwick Tree

tree[k] = sum_q (k - p(k) + 1, k)

Where, p(k) = k&-k and denotes the largest power of two that divides k

```
#include<stdio.h>
#include<math.h>

int sum(int arr[], int n, int a, int b){
    int i, sum=0;
    for (i = a; i<=b; i++)
```

```
        {
                sum += arr[i] ;
        }
        return sum;
}

void create_fenwick_tree(int T[], int arr[], int n)
{
        int a, b, k;
        for (k=1; k<=n; k++)
        {
                a = k – (k&-k) +1;
                b = k;
                T[k] = sum(arr,n,a,b);
        }

}

void display(int arr[], int n)
{
        int i;
        for (i=1;i<=n;i++)
        {
                printf("\t %d ",arr[i]);
        }
}
```

```
int main(){
	int n, i, arr[100], T[100] , a, b;

//	printf("the power of 6 is %u \n", largestPowerOf2(7)) ;
	printf("enter the number of elements\n");
	scanf("%d",&n);
	printf("enter the array elements\n");
	for (i=1;i<=n;i++)
	{
		scanf("%d",&arr[i]);
	}
	printf("\n Original array");
	display(arr, n);
	create_fenwick_tree(T, arr, n);
	printf("\n Fenwick Tree");
	display(T,n);
	return 0;
}
```

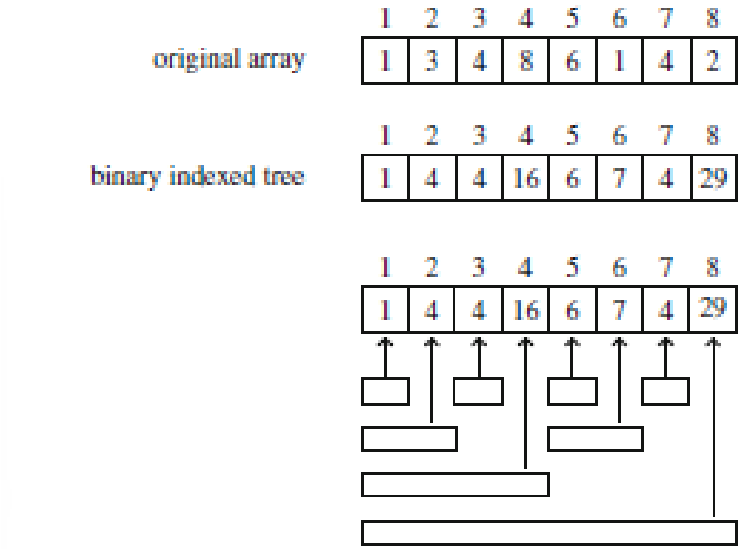
Fenwick Tree - Sum of Elements

Given a Fenwick Tree (Binary indexed Tree) with the value of at node k calculated as

tree[k] = sum_q (k - p(k) + 1, k)

Where, p(k) = k&-k and denotes the largest power of two that divides k.

Create a function to calculate sum of first ‘k’ elements in the Fenwick Tree.




```
#include<stdio.h>
#include<math.h>

int sum(int T[], int k) {
        int s = 0;
        while (k >= 1) {
                s += T[k];
                k -= k&-k;
        }
        printf("\n The sum is %d",s) ;
}
void display(int arr[], int n)
{
        int i;
        for (i=1;i<=n;i++)
```

```
        {
                printf("\t %d ",arr[i]);
        }
}
int main(){
        int n, k, T[100] = {0, 1, 4, 4, 16, 6, 7, 4, 29};
        printf("\n Fenwick Tree");
        display(T,n);
        printf("\n Enter the value for k");
        scanf("%d",&k);
          sum(T, k);
        return 0;
}
```